1. Compulsory Activities
(A) Need-based Activities

1. Introduction to Culture and the World of Work

1.1 Making a list of the industries in the surroundings and giving information about them

A flour mill : Milling various grains to make flours

Selling masks : Making and selling various kinds of masks

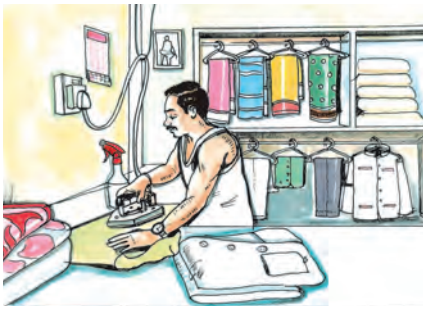
Provisions store : Retailing domestically required provisions and other articles

Photo-copy shop : Making photo-copies of documents

Fall and picot shop : Stitching falls and picoting *sari* edges

Tailoring shop : Stitching clothes for women, girls and children

Home delivered food service : Preparing and delivering meals to students, seniors and others who need them



Clothes-ironing service : Ironing clothes

My Activity

Visit the local Women's Self Help Group and obtain information about the industries and occupations seen there.

- Obtain information about industries other than those mentioned in the textbook.

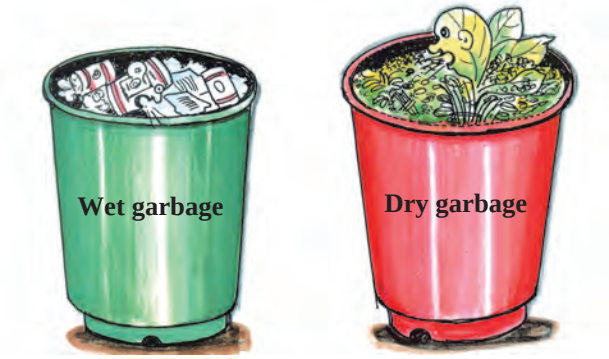
1.2 Importance of waste-management

Playlet

Hey, kids! Stop. You ate peanuts and left the skins and shells strewn in the classroom. Now see what a lot of rubbish there is all around you. Come on, pick it all up and put it in the dustbin.

What did you say? Who spoke?

Oho! I'm Rubbish speaking. Now listen, children, you must be wondering, there's so much rubbish all around, what can little children like us do about it? But, see, you need to think only about your school and your home. All the waste that collects in the house every day, can be separated into two parts, wet garbage and dry garbage. Stalks and skins of vegetables and fruits, spoilt veggies and fruits can all be used to make fertiliser by composting or vermiculture. Dry garbage like waste paper, wrappers, broken plastic or glass articles like dishes and bowls, bags, rags, boxes, cartons, bottles, old newspapers, flyers can be collected and handed over to garbage collectors or those who buy and resell such discarded items. Do you know the name for this process off separating and properly disposing off garbage? It is called Waste Management.



There are many advantages of proper waste management. Our surroundings remain clean. We do not get any foul smells. It prevents flies, mosquitoes and other insects from breeding. And that prevents pollution and disease. Thus it helps to maintain good health. And besides, we get fertilisers and we can enjoy the fruits and flowers from our gardens and farms.

Oh, very good! I see that you've cleaned up the classroom even as you were listening to my speech! Bravo, my dears!

Cleaning the classroom

My Activity

Imagine that you are speaking to the rubbish in your classroom. Write a speech of 5 lines.

- Give guidance regarding proper waste management, and disposal and about reuse and recycling.

1.3 Volunteering at local festivals, functions, celebrations

Our society loves to celebrate. Many festivals, fairs and functions are held in villages, towns and cities. To enjoy and to take part in them people gather together in large crowds. At such times, volunteers are needed to organize them well. A person who serves or helps people willingly and happily without expecting any return, reward or honour is called a volunteer.

Volunteers need to estimate the nature and number of people expected in the gathering for a festival or function and accordingly carry out many different tasks. Some of the tasks have been given below.

Spreading themselves out in different places in the crowd to control and guide people.

To help lost children to get to the proper places.

Looking after the footwear stand

Offering first aid

Preparing notice boards

Helping senior citizens

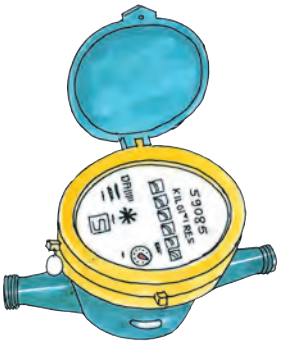
My Activity

Volunteer at a festival or other function and write a detailed letter to a friend describing your experience.

- Give guidance to children of different age groups to work as volunteers.

2. Water Literacy

2.1 Measuring water



Let us measure the amount of water used by one family in a day. There are several ways of measuring water.

(a) By using a water meter :

Tools and materials : water meter and bucket

Procedure

1. Place a bucket under a tap connected to a water meter. Open the tap and observe the water meter.
2. Using a bucket with a litre scale marked on it, note how much water collects in the bucket when the water meter shows 1 unit.
3. By noting the number of water units used up in a day and the amount of water that makes one unit of water, you can tell approximately how many litres of water are used in a day.

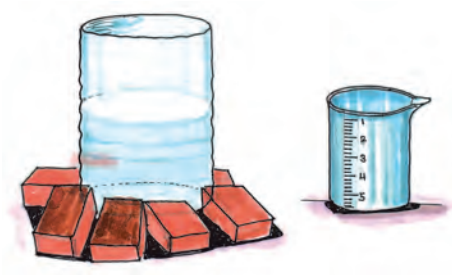
(b) Other methods of measuring water

1. Observe and take note of the capacity of the large tanks that supply water to towns/cities which is marked on them.
2. Water can be measured by observing the capacity marked on overhead tanks in our house.

2.2 Measuring rainfall

Rain is Nature's gift for all living things. It is important to measure rainfall to plan for future use of water. The laboratory of the meteorological department uses a rain gauge for this purpose. The height of the water column that collects in the measuring jar of the rain gauge gives us the amount of rainfall in 24 hours. Rainfall is measured in millimetres.

- **Let us make a rain gauge and measure rainfall**



Tools and materials : An empty water bottle, a serrated blade or knife, bricks or stones, a measuring cylinder or beaker

Procedure

1. Cut away the narrow upper portion of the bottle.
2. Before it starts raining, place the bottle in the open with the support of the bricks or stones.
3. When it stops raining, pour the water collected into a measuring jar and find its volume.
4. Record the rainfall at that time depending on the number of millimetres of water filled in the rain guage.

My Activity

Observe a rain gauge and draw a picture of it. Collect news items about weather and rainfall from newspapers and read them.

- Guide the students about how to use a water meter. Show them the water reservoirs in the village/ town/ city and explain to them their capacity. Give proper guidance when conducting the experiment for measuring rainfall.

2.3 Methods of transporting water

Let us see how water reaches our houses.

Rain water collects in dams, lakes, rivers and wells and we use this water for the whole year. So this water has to be brought to our village/ town/ city and to our houses and fields. Various kinds of methods are used to transport this water.

(1) **Carrying water on the head :** Water is carried in a pot, pitcher or other container supported on one's head.

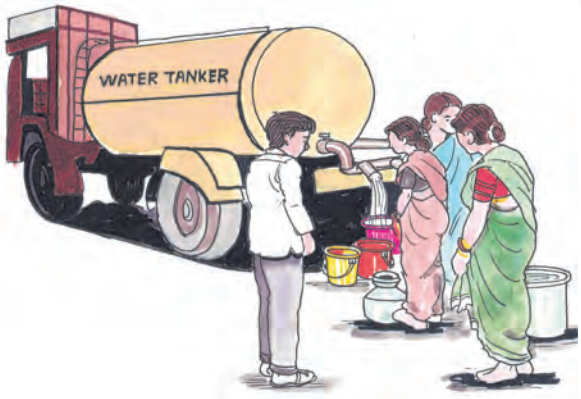
(2) **Carrying water in a waterskin :** The large wide leather bags of water hung one on either side of a pack-animal are called waterskins. They are mostly used in rural areas to carry water for use in homes.

(3) **Storing water using a water carrier:** A water carrier or *kaavad* consists of a long bamboo stick from which containers of water are hung by means of ropes.

(4) **Carrying water through pipes :** Pipes of varying sizes are used to transport water from the water source to every house. This is the most common method used everywhere nowadays.

(5) **Transporting water in water tankers**: A pump is used to fill water into tankers which are then driven to the parts where water is needed especially to drought hit areas in the summer.

(6) **Tube well :** Water that is deep underground is lifted to the surface through pipes by means of a hand pump or a motor driven pump.



My Activity

Write in short and draw a picture on the topic, 'This is how water comes to my house'. Make a model of a water carrier.

- Have a picture-reading and discussion or show a video or filmstrip introducing the children to the different methods of transporting water.

3. Disaster Management

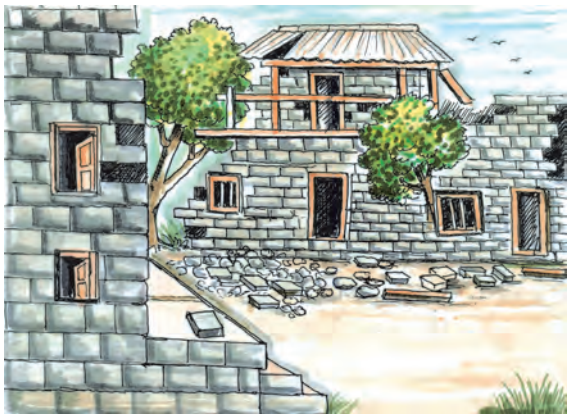
Natural Disasters

The crisis situations that arise due to the harm caused by certain natural phenomena are called natural disasters. Examples of such phenomena are storms, cyclones, floods, earthquakes, forest fires, tsunamis, epidemics, etc.

Sometimes we get a warning about its occurrence. Some steps can be taken to reduce the harmful effects of the disaster and the damage caused by it to life and property.

The plan put in place to reduce the severity of the effects of a natural disaster is called 'natural disaster management'.

• Earthquake : Movements that take place deep inside the earth release enormous amounts of energy. They result in rapid movements under the surface of the earth in the earth's crust. The surface of the earth also moves. The ground under us shakes, tremors occur and cracks appear in it. Such a shaking of the earths surface for a few moments is called an earthquake.



• Causes of an earthquake : When layers of rock or tectonic plates slide over one another beneath faults in the earth's crust, then sudden movements take place and an earthquake occurs. Chemical reactions that take place between the liquid and gaseous constituents of the earth's mantle/ core can cause an earthquake. An explosion in a mine or one caused by any other human activity can also cause an earthquake.

The tremors caused by an earthquake may be mild or intense. Their intensity is measured on a scale called the Richter scale.

• Effects of an earthquake : Houses, buildings collapse, trees are uprooted.

• Measures for minimizing the damage caused by an earthquake :

(1) When you feel the tremors, leave the building and stay out in the open.

(2) Crouch under a strong table, bed or bench.

(3) Fold a thick sheet/ blanket/ quilt many times over and protect your head with it before going outdoors.

(4) Turn off the gas cylinder/ stove in your house.

(5) Switch off all electrical appliances.

(6) Stay far from electric supply lines, lights, bridges, buildings.

(7) If driving in a vehicle stop in a safe place. Avoid stepping out.

(8) Help each other to keep your morale high.

(9) Face the crisis with confidence. Do not panic.

• **Floods :** Heavy rains cause water levels to rise and that causes floods.

• **Causes of flood :** Encroachments into river and nullah beds, increasing amounts of dumped garbage make the river/ nullah beds shallow and that causes flooding. An earthquake may cause a crack in a dam wall and give rise to the possibility of a flood. Water level rises during a high tide and causes flooding. Barrages, weirs may also get damaged resulting in floods. Meltwaters from the Himalayas may also cause floods.

• **Measures for reducing damage caused by a flood :** Houses should be constructed at safe distances from river banks. Plant trees. Protect and conserve forests. Desilt reservoirs of water regularly. Remove the silt and other garbage from the beds of rivers, streams, lakes and dams before the start of the monsoon season. Organize teams of local doctors and volunteers and ensure that hospitals and health centres are functional. Give advance warnings to people living near the waters and evacuate them if necessary. Take necessary action to prevent outbreak of diseases.

• **The Corona Virus (Covid 19)**

This is an infectious diseases caused by the SARs-CoV-2 virus. We can get the infection if we breathe in the virus present in the air exhaled by an infected person while we are near them, or, if we touch a contaminated surface with our hands and with the same hands touch our eyes, nose or mouth. This virus spreads very easily in the house and at the crowded places. We should take proper care to protect ourselves from Corona.

• **Take all proper precautions to protect yourself from Corona, namely :**

• Frequently wash your hands clean with soap and water.
• Maintain social distances.
• Avoid touching your eyes, nose and mouth.
• Clean your hands with a sanitizer.
• Cover your mouth with a tissue or handkerchief while coughing or sneezing.
• Use a mask to cover your nose and mouth.

My Activity

Collect slogans relating to a disaster like an earthquake, flood or COVID19. Collect clippings of news items or pictures of a cyclone or tsunami. Make a list of the aid that you can give to those affected by a flood.

◆ Give information about the precautions to be taken before, during or after the occurrence of an earthquake, flood or epidemic.

(B) Activities of Interest

1. Making a hat from old newspapers

Tools and materials : Old newspapers, large sized coloured paper, a big colourful wall calendar, other decorative items, scissors, glue, etc.

Procedure

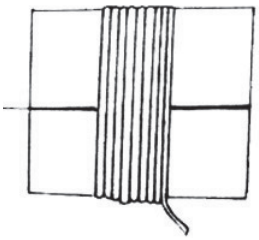
1. Take the two joint pages of a newspaper or a large sheet of coloured paper folded at the centre.
2. Find the centre point of the long folded side of the paper. Now lift the corners of this closed side and fold the paper inwards till the centre point.
3. Now press the folds into place. From here onwards follow the actions shown in the pictures below to finish making the hat.
4. Flip the paper over and fold both the corners inwards.
5. Fold the extra strip below, upwards and fix it in place.
6. Complete the hat as shown in the pictures below.

My Activity

Make a variety of hats using old newspapers and gift them to your friends. Write out the procedure for making a hat and ask your friend to make a hat by following those instructions.

- Arrange an exhibition of the decorative hats made by the children.

2. Making tassels and decorative hangings using strings, threads and beads



Tools and materials : String, beads of various sizes, shapes and colours, a thick paper dish or round piece of cardboard, thick thread, an old comb, the old gasket of a pressure cooker or other large ring, bangles, needle, scissors, glue, etc.

Procedure 1

(A) Making the tassels

1. Take square cardboard piece of 4 × 4 cm size. Wrap the thick thread around it 10 to 12 times or more if necessary.
2. Pull the wrapped thread up from both ends and tie the loops together tightly.
3. Separate the loops of thread from the cardboard piece, tie it and cut it as shown in the picture.
4. Use an old comb to comb out the fibres of the threads in the loops to make a pretty fluffy tassel.
5. Make tassels of various sizes by the same method.
6. Tassels can be made from woolen yarn of various colours instead of string/ thread.

Procedure 2

(B) Using tassels, beads and thread to make a decorative hanging

1. Wrap string/ thread around the gasket and the bangles and cover them.
2. String some beads on a thread and decorate the middle of the bangle or gasket.
3. String together the beads and tassels to make it look decorative. Make pairs of such decorative strings of varied lengths. Hang them from the decorated ring or gasket in the order of their lengths.
4. Tie strings on the upper side of the ring, bring them together to make a hook for hanging.
5. Use mirrors and sequins to decorate your hanging further.

My Activity

Make various kinds of tassels and hangings.

Draw an artistic picture and decorate it with beads and string.

- Give proper guidance to the children as they make the tassels and hangings.

(C) Skill-based Activities

1. A lantern made by folding paper

Tools and materials : 2 sheets of A4 sized (21cm×30cm) coloured paper, thick thread, needle, scissors, glue, etc.

Procedure

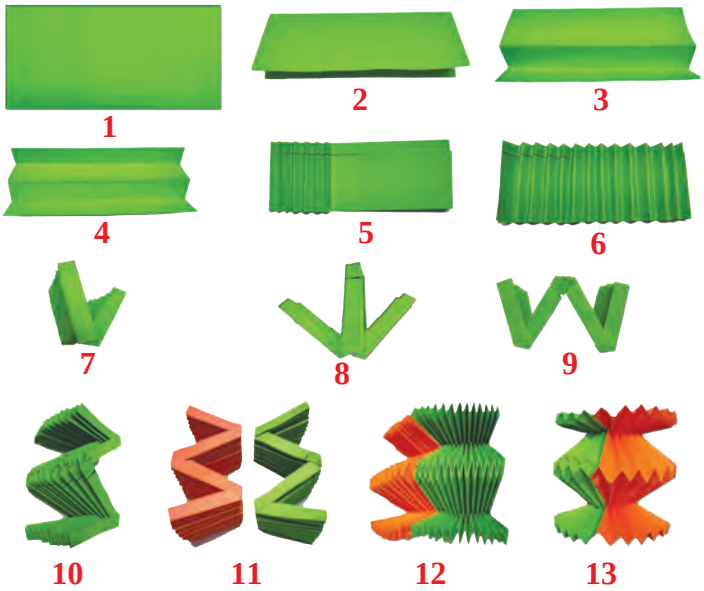
(1) Take the two sheets of paper described above and fold them into a 'W' shape following the figures 1 to 4 below.

(2) As shown in figure 5, starting from the narrow side, make small folds in the paper at equal distances like a fan, till you reach the other end. You will get an 'I' shape. Press the folds down if necessary to make them tidy.

(3) Now fold this 'I' shape to get a 'V' shape. Now fold the other sides to get a 'W' shape. You will see from the figure 10 that one side of the lantern is ready. Make the other side also in the same way.

(4) Now open out these two parts and give them a rounded shape. Stick the two parts to each other. Tie 3 threads at equal distances. (See figure.)

(5) Tie narrow strips of golden or silver paper beneath the lamp to decorate it. Hang your lantern and display it.



My Activity

Try making paper lanterns of different colours.

- Guide the children to make the folds properly.

2. Making greeting cards using straw, leaves, flowers

Tools and materials : Coloured cardsheet, dried grass, leaves, dried flowers of different colours, glue, sketch pens, scissors, etc.

Procedure

(1) Collect flowers, leaves, grass from your surroundings. Place them between two pieces of newspaper and put them under some heavy books that are not being used. (A herbarium sheet or blotting paper can also be used.) In about 2 to 4 days they will have got pressed, flat and dry.

(2) Cut the cardsheet into a suitable size for your greeting card. On the smooth right side of the card paper draw a border using a sketch pen.

(3) Take some of the dried flowers leaves, etc. of the right size to make an attractive pattern that will fit inside the border you have drawn.

(3) On the greeting card, first stick the straw and grass and then the leaves and flowers on top of them. Decorate the card as you wish.

My Activity

Stick dried grass, leaves and flowers on a paper to make a birds' nest picture.

- Show children how to paint a picture on the mesh of the dried leaf of a peepal tree.

2. Optional Activities
(A) Productive Fields of Work

1. Area : Food

1.1 Kitchen Gardening

Gardening tools and their uses

Pickaxe : for digging and loosening the soil

Hoe : for digging and loosening the soil

Spade : for drawing soil together, making the ground even, lifting soil into a basin

Basin : for carrying soil or manure from one place to another

Gardening trowel : for digging the soil for weeding

Digger : for making pits or furrows in the soil for transplanting seedlings

Hand fork : loosening soil for picking out stones

Fork : for prising out seedlings for transplantation

Secateur : hand tool for pruning branches

Watering can : for watering small plants

Spray pump : for spraying insecticides/ germicides on plants

My Activity

Write what tool you would use for the following purposes.

1. To carry manure for plants .....
2. To water potted plants .....
3. To prepare a bed .....
4. To spray a pesticide on vegetables .....
5. For weeding .....

- Give children the opportunity to use tools as and when the need arises for using them.
- Explain the precautions to be taken while using the tools, well in advance. Give guidance where necessary.

1.2 Simple Farming

Introduction to different types of crops

Our meals include a variety of preparations like *daals*, rice, *chapattis*, *bhakaris*, *usal*, jaggery, sugar etc. We also use oil. The grains, pulses, sugarcane, oilseeds required for these preparations are grown in fields and are called crops. Let us learn about the different types of crops.

Seeds must be sown in the fields to grow crops.

Seeds of different crops are sown at different times of the year. Only when seeds are sown at the right time and climate do we get a good crop. The period of the year when a crop grows best its called its season. According to their seasons, crops are divided into three types, namely kharip, rabi and summer crops.

Crops can also be divided into the following four types

1. Cereals
2. Pulses
3. Oilseeds
4. Cash crops

1. **Cereals :** Jowar, wheat, bajara, rice are cereal crops.
2. **Pulse crops :** Green gram, Bengal gram, black eyed beans, moth bean, broad bean, and other kinds of beans, peas are pulses.
3. **Oilseed crops :** Oil is extracted from groundnuts, and from seeds of sesame, flax, safflower, sunflower, castor, mustard and soya. Therefore they are called oilseed crops.
4. **Cash crops :** Crops like cotton and sugarcane whose produce is sold for cash directly after harvesting are called cash crops.

Besides the above types there are also other types of crops such as fodder crops, horticultural crops and forest crops.

My Activity

1. Write the answers to the following questions.

i. Name the main types of crops.

ii. Of which type is each of the following crops?

(a) wheat (b) sesame (c) green gram (d) fenugreek (e) sugarcane

iii. What is called the specific time for sowing?

2. Visit an oil mill to observe how oil is pressed out of oil seeds and write a description of the process.

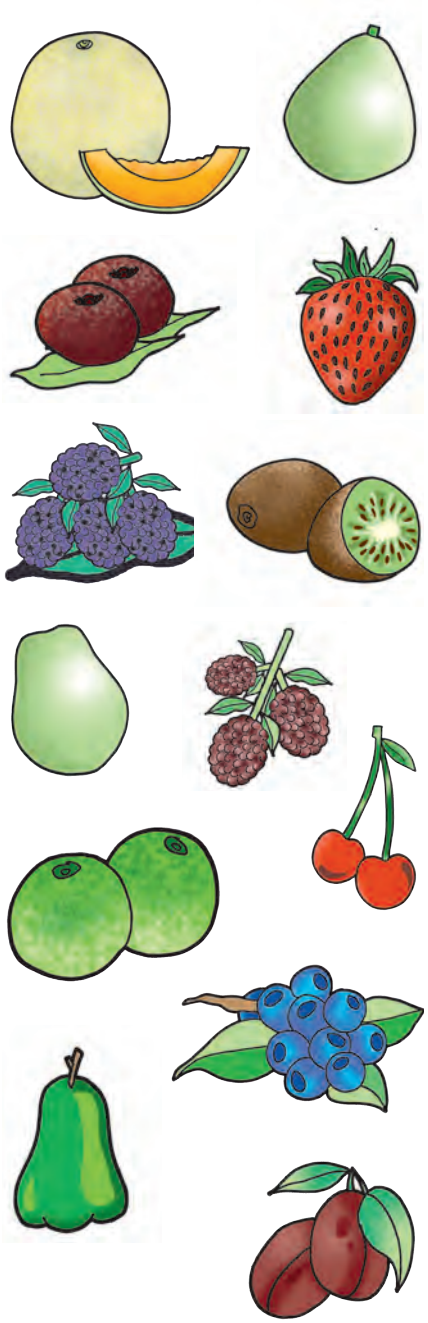
- Help the children to collect specimens of some crops.

1.3 Fruit Processing

Getting to know some unfamiliar fruits

Each fruit requires a specific kind of climate to grow. That is why we find that all fruits do not grow in our region. Besides, some fruits ripen only in a certain season. They are not available throughout the year. We may have heard their names but we are not easily seen in our surroundings.

Sr. No.	Name of fruit	Colour	Shape	Taste
1	Gooseberry	Orange	Round	Sweet and sour
2	Muskmelon	Greenish peach	Round, rough outside	Sweet
3	Pomelo	Greenish	Round, oval shaped	Sweet and sour
4	Kokum	Deep red	Round	Sour
5	Strawberry	Reddish	Small round tapering on one side	Sweet and sour
6	Mulberry	Black	Round	Sour
7	Kiwi	Rough, chocolate brown	Oval	Sweet
8	Pear	Greenish, white	Oval	Sweet
9	Lichi	Light brown	Round	Sweet
10	Apricots	Yellow	Round	Sweet
11	Wax-apple / Rose- apple	Green, pink	Round, tapering	Sweet
12	Cherry	Red	Oval	Sweet
13	Wood apple	Greenish	Round	Sweet and sour
14	Blueberry	Deep blue	Round	Sweet and sour
15	Bael/Bell	Greenish	Round, oval	Sweetish
16	Plum	Red, blackish red	Round	Sweet and sour



My Activity

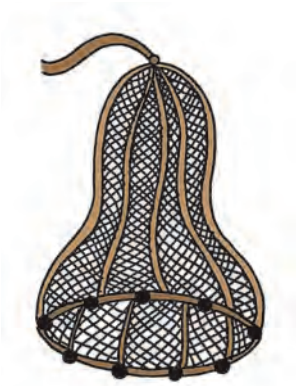
Write the names and uses of some of the unfamiliar fruits.

- If possible, show children these exotic fruits and give some information about them.

1.4 Pisciculture

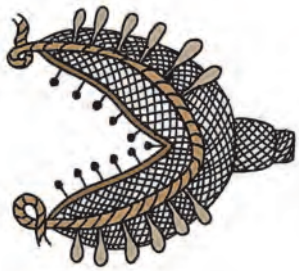
Introduction to the types of nets used for catching fish

Various kinds of fishing nets are used, according to necessity, to catch fish. The type of net is chosen depending upon the place where fishing is to be done.



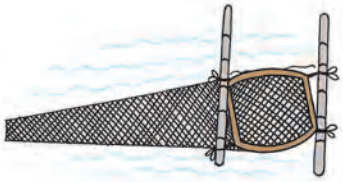
Cast net

Cast net : This net is round in shape like a cone, tapering to one end and round and open at the other. A 2 to 3 metre long rope is tied to the tapering closed end. Iron or lead shot are tied to the lower border and this part is turned in and stitched up. Standing at a height and holding on to the rope at the tapering end with one hand, the net is thrown into the water with the other. This net is used in shallower waters.



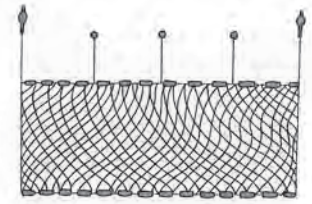
Drag net

Drag net : This is a very long net. Its mesh size is 2.5 to 4 cm and the width of the net is 3 to 4 metres. A nylon rope is woven into the upper border of the net and floats are attached to it at regular intervals along its length. A thick nylon rope is woven through the lower edge of the net and lead or iron weights are hooked to it. The net is pulled by both ends across a river channel and brought to the river bank. This requires a lot of manpower.



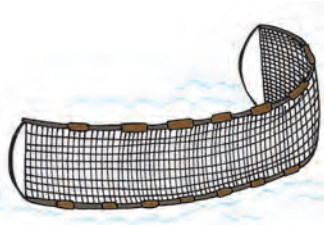
Bag net

Bag net : A bag net is triangular in shape. Its length may be from 15 to 200 metres. Its mesh size is 5 to 10 cm. This net is fixed in one place across the direction of the current in the sea.



Gill net

Gill net : Nylon or cotton yarn is used to make this net. Floats are attached to the upper line of the net and weights to the bottom line. The net is kept, fixed vertically like a wall in the water all night. Fish swimming in the water get caught in the net.



Shore seine net

Shore seine net : This is the biggest type of net. Its width is 400 to 500 metres and its height, 10 to 12 metres. The mesh size near the centre is 3 to 5 cm. Floats are attached to the top line of the net and lead weights to the bottom line. A 300 to 400 metres long and 7.5 cm thick rope is tied to both ends of the net. 20 to 25 people are needed to pull this net.

My Activity

If possible, watch in person, how net fishing is carried out.

- Explain the precautions that need to be taken while fishing.

1.5 Food Production

Tools required in food production

All kinds of dishes are prepared for eating in every house, restaurant or big industries in smaller or bigger quantities. Let us get to know the tools and devices used in food production.

Sieve : To separate small stones and other rubbish from flours and grains.

Cutting board : To cut fruits and vegetables.

Peeler : To remove the peels of fruits and vegetables like potatoes, gourds.

Knife : To cut vegetables and fruits.

Colander : (Made of cane/ stainless steel/ plastic) To drain the water from fruits and vegetables, etc. after they have been washed.

Kitchen tongs : To lift hot vessels. Kitchen tongs are of two or three types.

Mortar and pestle : Made of steel, stone or brass used for pounding.

Grater : To grate coconut, carrots, radishes, cucumber. There is a special one for nutmeg.

My Activity

All the above implements are manually operated. However, nowadays electrical appliances are used in food production. Make a list of these appliances and make a chart like the one above, showing their names and uses. Collect and stick their pictures in a scrapbook.

- Give proper guidance for making the list and the chart.

A bhel of mixed pulses

Ingredients and their proportions : ¼ cup each of various sprouted pulses such as green gram, moth, etc., 1 tablespoon horse gram,1 tablespoon tamarind and jaggery chutney, 1 teaspoon ground green chilly, 1 onion and 1 tomato finely chopped, ½ teaspoon chaat masala, salt to taste and for garnishing 1 spoon shev and ¼ cup chopped coriander greens.

Procedure

1. Wash the pulses and leave them to sprout.
2. Steam the sprouted pulses lightly. Remove the steamed pulses in a glass or steel container.
3. Add the other ingredients and mix well. Serve into bowls and garnish with shev and coriander greens.

My Activity

Write down the process of sprouting the pulses for making bhel.

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- Explain the advantages of eating sprouted pulses.

2. Area : Clothing

2.1 Producing Cloth

Yarn is drawn from cotton, wool or silk and is used to produce cloth. This process is called 'weaving'. In olden times cloth was woven by hand. Later the spinning wheel was invented. Of all the devices and machines needed in cloth production, the spinning wheel or *charkha* comes first.

The *charkha* evolved very rapidly. There are many types of *charkhas*. Let us get to know the most important of them.

1. **Khada *charkha* or standing/ upright spinning wheel :** This *charkha* has a wheel of 40 to 60 cm and a disc. The wheel is connected to the disc by a cotton belt. How to spin a yarn with the help of a spinning wheel is still taught at the Sevagram Ashram.

2. **Kisan *charkha* :** The main parts of this *charkha* are the following : Main wheel, speed, wheel thin string, thick string, spring, disc. This *charkha* has two wheels. The main wheel of 20cm diameter while the smaller wheel which helps to increase the speed of the *charkha* has a 10cm diameter. While spinning yarn the main wheel is parallel to the ground. This *charkha* spins more yarn than the khada *charkha*. This *charkha* is also called 'Yervada *charkha*.'

3. **Peti or box *charkha* :** This *charkha* is available in various sizes and is fitted in a box. It can be opened or shut with the help of hinges.

4. **Pravasi *charkha* :** This is a portable *charkha* which can be carried along even while travelling. But the speed of this *charkha* is very less.

5. **Ambar *charkha* :** This is the newest version of the *charkha*. This has a wooden spinning wheel which can be worked by hand, 4 spindles and 3 pulleys. It has 2, 3 or 4 grooves and each pulley is connected to the main wheel by means of a cotton belt.

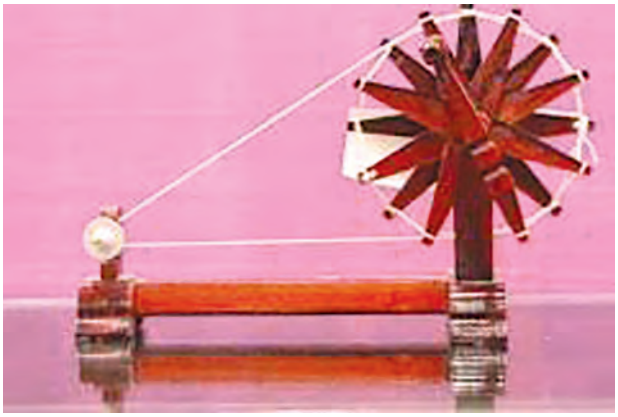
   It has an automatic device for wrapping the yarn. Although the yarn obtained using the ambar *charkha* is rough and of ordinary quality it spins four times as much yarn as any of the other *charkhas*.

   The importance that Gandhiji gave to the *charkha* as a village industry still continues.

My Activity

Obtain information about *charkhas* by visiting a *charkha* museum or exhibition.
Obtain more detailed information about one of the *charkhas*.

- Give information about weaving equipment using a *charkha* or pictures/ videos of a *charkhas*.



Khada *charkha*

Ambar *charkha*

Kisan *charkha*

Peti *charkha*

2.2 Basic Tailoring

Stitching a hook on to a garment

We wear many different kinds of dresses. They are made to fit us well, and easy to wear and to take off by the use of certain simple devices such as drawstring, hooks, loops, buttons, etc. Of these the hook is a very commonly used device.

Tools and materials : Cloth, hook and eye, needle, thread, etc.

Procedure :

1. Thread a needle. Bring the two ends of the thread together and tie them in a knot.
2. Place the hook on the cloth in such a way that its round loops are towards the centre of the cloth and the head of the hook is at the edge of the cloth.
3. Pass the needle through the cloth and the loop of the hook so the knot is on the wrong side of the cloth.
4. Now pass the needle back to the wrong side from a point close to the hook's loop but outside it.
5. Repeat these two steps so that there will be several stitches around the hook's loop.
6. Now bring the thread up though the other loop of the hook and repeat the same two steps.
7. Next, bring the thread up one side of the top of the hook and pass it down from the other side thus holding down the head of the hook. Put in 2 or 3 stitches like this, side by side rather than all in the same place.
8. In the end, before pulling the thread tight after the last stitch, pass it through the loop formed by the loose thread and thus tie a knot. Then cut the thread with scissors.

My Activity

Observe an old dress which has hooks. Replace a missing hook by the method described above.

- Give guidance where necessary.

2.3 Doll Making

Making a doll from waste materials

Tools and materials : Old plastic bottle, toothpick/ stick, black/ dark brown woolen yarn, old newspaper, coloured paper, glue, decorative baubles

Procedure :

1. Take a medium sized plastic bottle. With the help of an elder person make a hole in the lid of the bottle using scissors points or a nail. Fix a toothpick or any stick in this hole.
2. Take a sheet of coloured paper and fold it into pleats. Stick it in such a way as to make a long skirt for the doll. Stick another paper on the upper portion of the bottle including the lid to make a blouse.
3. Make a small ball using clay, plasticine or crushed paper, or use a table tennis ball to make the head.
4. Stick the black or brown strands of wool on the head to make hair. Draw the nose, eyes, and lips in the proper colours.
5. Roll paper and flatten the roll to make the dolls arms and attach them.
6. Decorate your doll with attractive baubles like sequins, beads, lace and shiny coloured paper.
7. The lid of the bottle is the doll's neck. Remove the lid and fill the bottle with small, dry sweets.
8. Your doll is both decorative as well as useful from waste materials.

My Activity

Make a doll using a coconut shell.

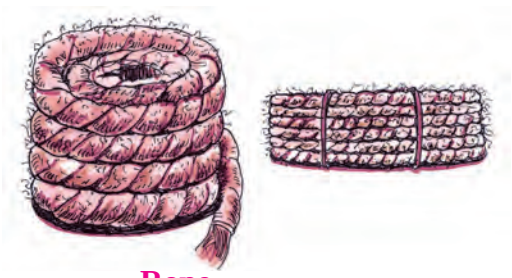
- Give guidance where necessary.

2.4 Coir Work

Various articles made using coir

When you make a list of the things made of coir, remember to classify them to make it easier to make the list.

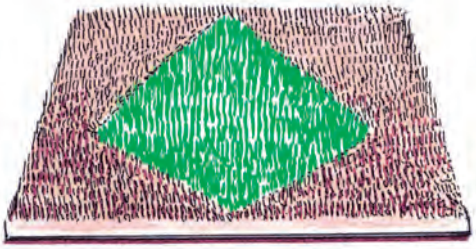
1. **Articles of household use :** Rope, string, pot hanger, doormat, wall hanging, decorative articles.
2. **Articles of use in the fields :** Rope, sling
3. **Articles for sale :** Carpet, swing, mattresses, pillows
4. **Other common articles :** Bag, brush



Rope

String

Wall hanging



Doormat

Pot hanger

My Activity

Make a list of the coir articles available in your house.

- Urge the children to recall the coir articles they have seen, to name them and to make a classified list of those articles. Help them if necessary to do the classification.

3. Area : Shelter

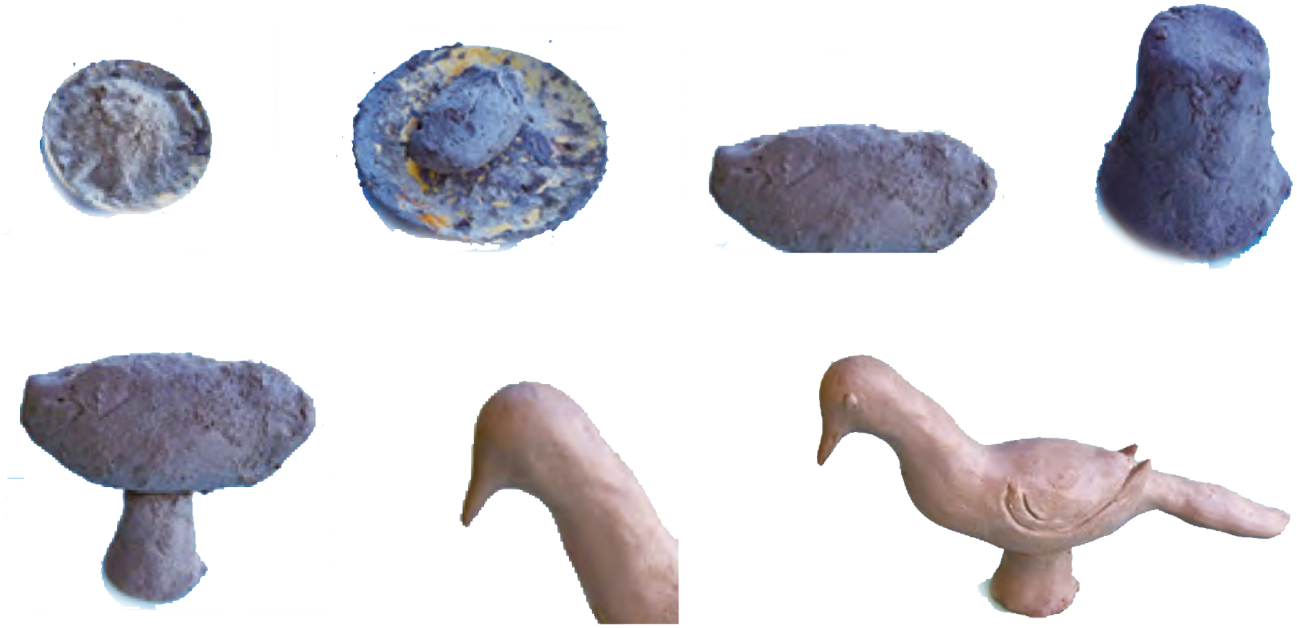
3.1 Clay Modelling

A clay bird

Tools and materials : Clay/ Clay like soil

Procedure :

1. Take some clay. Add enough water and knead it so that it can be shaped into a ball.
2. As shown in the picture shape some clay into an upright stand. Also make the shape of the body.
3. Fix a pin or stick on the top of the base and fit the body properly on top of the base. Prepare the shapes of bird's neck, head and beak.
4. When fitting the bird's head to the body use a thick wire or stick to give support. Make small clay balls for eyes or use beads. Shape some clay into bird's wings.
5. Thus you will get a clay bird.



My Activity

Make a clay model of a tortoise.

- Offer help and guidance wherever needed.

3.2 Bamboo and Cane Work

Planting, harvesting and storing bamboo

Bamboo can be grown in any kind of soil from the most fertile kind to red, yellow, *murbad*, light, shallow, clayey or sandy soil. Bamboo may be planted on bunds between fields, open spaces around houses, river banks or even hill slopes.

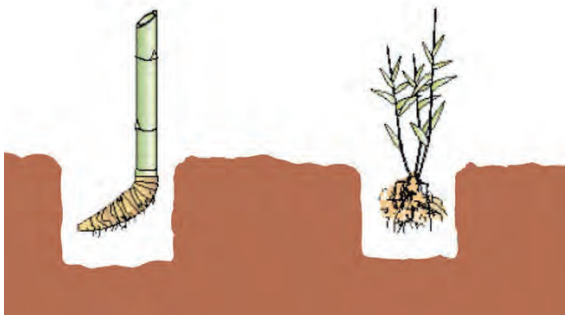
Bamboo may be grown by either of the two methods, namely vegetative or from seeds.

1. Vegetative method : This is the traditional method. Bamboo is grown from an already existing bamboo grove. There are some sub-types of this method.

i. Growing the small variety of bamboo having a runner stem.

ii. Planting one year old culms.

iii. 1 metre long shoots of the bamboo are planted at a slant to the ground.



2. Sowing a seed : Planting a bamboo by sowing a seed in the ground.

Method of planting : A pit of 30cm × 30cm × 30cm size is dug and the bamboo is planted in the month of June. A distance of 1.5 m between 2 plants and 3 m between two rows is maintained.

Harvesting the bamboo : Bamboo is generally harvested in the months of October or November. The bamboo is cut about 20 to 30cm above the ground. Three to four year old bamboo sticks are quite durable.

Storing bamboo : After cutting, the bamboo sticks are kept upside down. They are soaked in a pesticide. They are kept under running water for a month and then dried in the shade.

My Activity

Visit a bamboo plantation. Observe how the work is carried out there and write a description of it. Study how bamboo is grown in your surroundings.

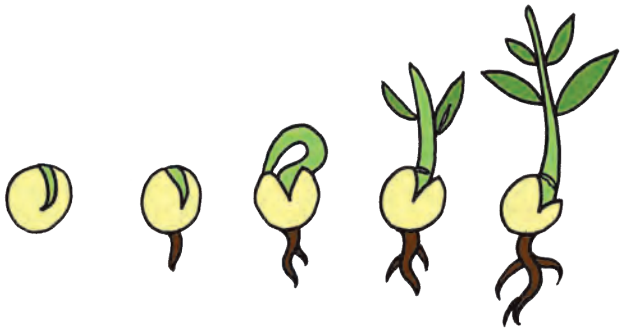
- Offer help and guidance to the students wherever needed.
- Arrange a visit to a bamboo farm and give students an opportunity for observation.

3.3 Cultivating Flowering and other Ornamental Plants

Unit : Collecting seeds of flowering plants according to their season and preparing seedlings in the proper way

Dear friends! Flowering and other ornamental plants make our surroundings beautiful, isn't that so ? So let us prepare seedlings of some flowering plants.

1. Collect the seeds of beautiful flowering plants like marigolds, asters, chrysanthemums, babul, gulmohur, bahava, etc. in their season, when they are available. Clean and dry them. Then store them in plastic or paper bags. On each bag, write the name of the plant and the date on which the seeds collected.
2. To prepare the seedlings, fill a small pot or plastic bag with a mixture of soil and organic fertilizer. Sow the seed in it and water it.



My Activity

Sow the seeds and grow the seedlings. Observe the pot every day and write how it grows.

- Give proper guidance regarding collecting seeds, storing them safely, preparing the seedlings watering them, etc.

(B) Other Areas

Animal Husbandry and Poultry
(Conservation of Animals and Birds)

Use of productive domestic animals other than for milk

Certain occupations, if carried on along with agriculture, can help increase income. These are called agribusinesses or complementary occupations to farming.

Dairy : Milk is a part of the diet of all people from newborn babies to the aged. Milk is easily digested and is rich in salts. It is the best balanced diet for small children. That is why it is said to be a complete food. Milk is processed to make several other food stuffs, for example, yoghurt, buttermilk, *lassi*, *ghee*, *chakka* (hung curd), *shrikhand*, etc.

Milk products made by evaporating the water from milk : Ice cream, milkshake, *basundi*, *rabdi*, *khoya*, *pedha*, *kalakand*, *barfi*, *kheer*, *kulfi*, *gulab jamun*, etc.

Milk products made by curdling milk (processing milk by adding an acidic substance) : Cheese, *paneer*, *rasgolla*, *rasmalai*, etc.

Yoghurt : Heat the milk till it boils and allow it to cool to 30 to 35 degrees C. Add some drops of lemon juice or some yoghurt to it and stir. Cover it and keep it aside. The yoghurt will be ready in 10 to 12 hours. A culture of lactic acid bacteria is used to make yoghurt in large quantities.

***Chakka* (hung curd) :** Take good quality yoghurt and tie it in a thin cotton cloth. Allow all the water to drain from it. What is left in the cloth is hung curd or *chakka*. Ten litres of buffalo milk gives 3 to 3.5 kg of *chakka*.

- **Uses of domestic animals other than for milk**

Camel : The camel is an animal of the desert regions. A camel is used for transport. That is why it is called the Ship of the Desert. A camel is also kept for its milk and meat. Camel hide is used to make footwear and its hair is used for certain industrial clothing.

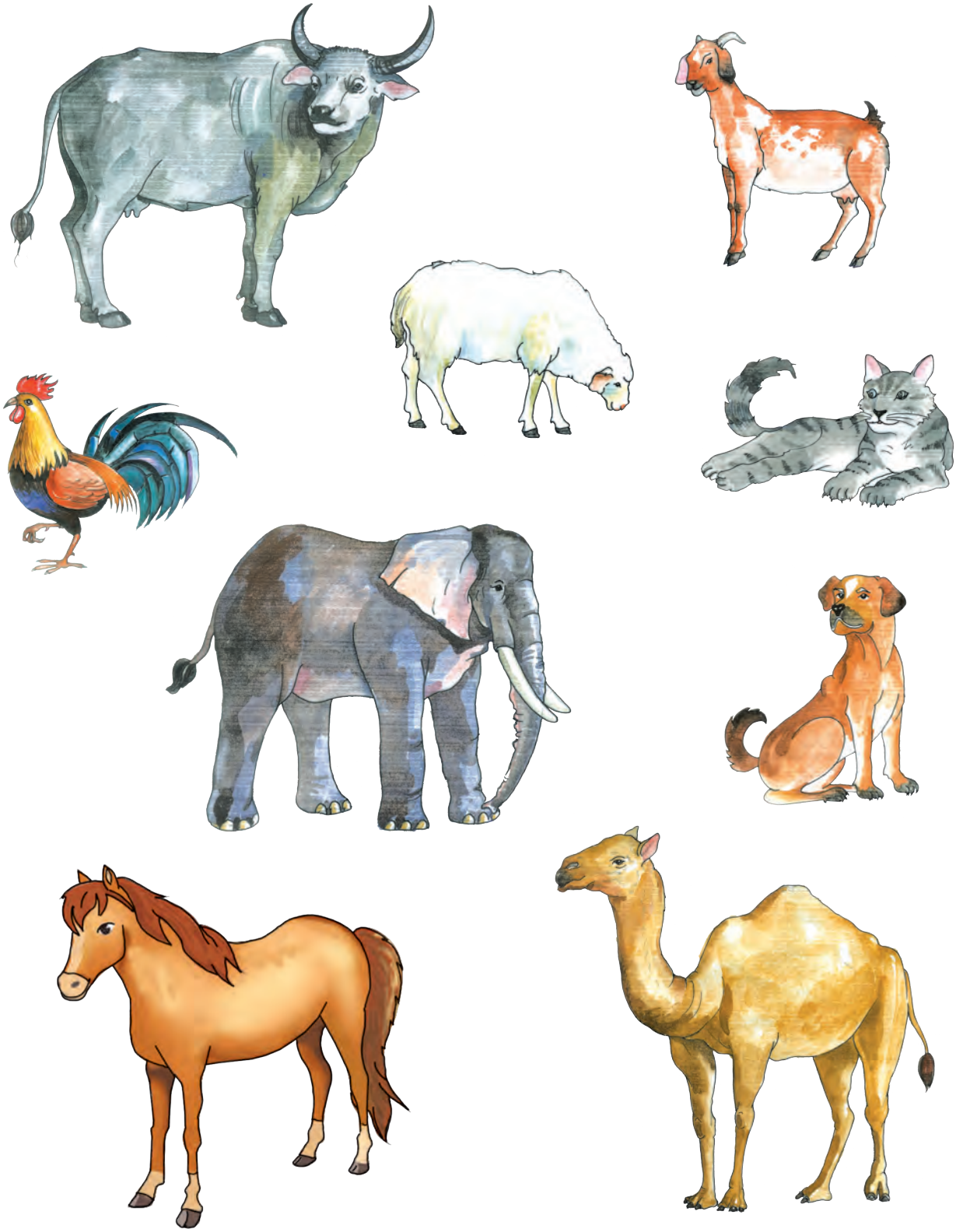
Yak : The yak is an animal found in the Himalayan region. It is kept for use in agriculture, transport and for other heavy work as well as for meat. Clothes, toys, dyes/ colours, ornamental articles are made from its hair.

Pig : The pig is to be found everywhere. Pigs/ boars are kept for meat. It is called pig keeping or pig farmimg. Pigs are reared for their meat as well as hide. The skin is used to make belts, shoes, jackets, gloves, the hair for making brushes and their fat for soaps, candles and lubricants.

- If possible arrange a visit to a nearby agribusiness centre such as a poultry or sheep farm.

My Activity

Write the name and some information about each of the animals shown in the pictures below.



- Explain the uses of domestic animals according to their uses at the local level.

(C) Area : Technology

(1) Information and Communications Technology (ICT)

Uses of the computer :

1. **Advertising :** A computer is used for all kinds of advertisements. It is used to prepare advertisements that appear in newspapers and magazines by filling colours and making them more attractive.
2. **Printing :** A computer is useful in all kinds of printing processes such as newspapers, magazines, periodicals, reports, books, etc.
3. **Medicine :** In the field of medicine, computers are used for conducting tests to obtain a complete picture of the patient's condition to record the current condition of the patient with respect to blood pressure, body temperature, pulse rate, etc. Today computers are also used to help with complex surgeries.
4. **Education :** In the field of education computers are very useful in making the teaching and learning process more effective. Learning and appearing for an examination have become easier with the help of the computer. Computers are also useful for preparing the exam results of students as also for keeping records of the general register.
5. **Banking :** In banks, computers are used to keep a record of the transactions in customer accounts. In some cities, an identity card can be used for financial transactions through the ATM or the Automatic Teller Machine. This stored information can be processed to obtain a solution. This is called Data Storage.
6. **Post and Telephones :** Computers are used to a great extent in the post and telegraph field, for example, for correspondence, sending messages, preparing bills, for controlling, etc.
7. **Personal uses :** We use computers to get information and scientific knowledge as also for entertainment. We use the Internet for online shopping, reading newspapers, obtaining information about various kinds of things from the websites of the relevant companies, referring to encyclopaedias online, etc.
8. **Agriculture :** In the field of agriculture, computers can help to increase production and to expand agribusinesses by providing information about, for example, agricultural research, studies of soil, weather and climate, and market rates of produce in various market places.
9. **Police :** Computers help to expose crimes. Computers can be used to keep records of stolen articles, vehicles and notes of an accident or crime scene as also finger prints, facial features, results of tests of mental balance, and information about lost people.

My Activity

Visit a place in the neighbourhood where computers are being used.

The use of tools in Paint

Atharva made the picture you see alongside, using the computer application called Paint.

Paint is one of the various applications which we can use for doing specific tasks with the help of the operating system of a computer. Using the Paint application we can make many different pictures, just like Atharva did. Paint offers all such tools as an artist needs while painting a picture on paper, such as a brush, colours, eraser, etc.

Let us see how Atharva made the picture above.

- He opened the Paint application.
- He used the Shape tool to select the triangle shape.
- He used the Fill Colour tool to fill it with the colour of his choice.
- Next he used the Shape tool again to select a rectangle.
- Then he used the Fill Colour tool again to fill the colour of his choice.
- And last, he Saved his File.

My Activity

Drawing a picture using various geometrical shapes. Saving the picture or file.

- Give the children the opportunity to use the Paint application using an available open-source Paint application.

Road Safety

Give Way

No Entry

One Way Traffic (as shown by arrows)

One Way Traffic (as shown by arrows)

Public Telephone

Petrol Pump

Hospital

First Aid Post

Left Trun

Right Turn

Right Hairpin Bend (Turn)

Right Reverse Bend (Turn)

Left Turn Only

Compulsory Ahead Only

Compulsory Ahead or Turn Left

Blow Horn

My Activity

Explain to the class about 'The use of GPS Technology to me'.

1. Picture

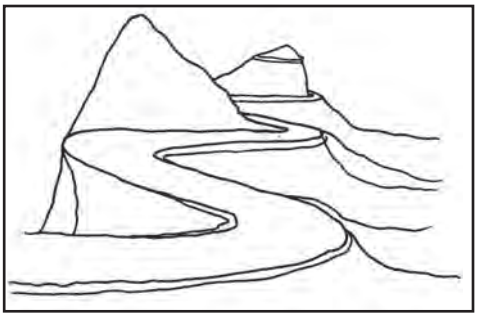
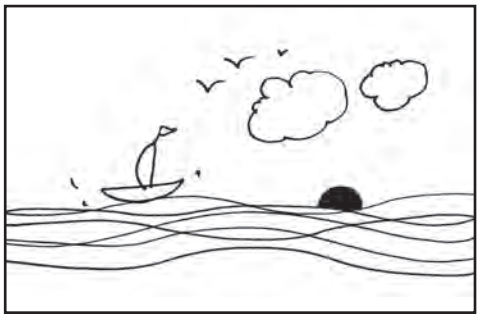
1. Sketching

1.1 Lines

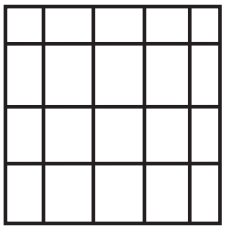
It is necessary to practice lines in order to draw pictures. Use different mediums like colour pencils, crayons, coloured pens or sketch pens to practise lines.

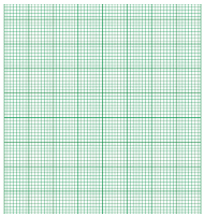
We have practiced vertical lines, horizontal lines, slanting lines, circular lines, zigzag lines and intertwined lines in the earlier Standard. Let's now practise two more types of lines.

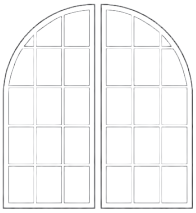
Wavy lines: The curving lines are called wavy lines. We find curving lines on waves on water, roads in mountain pass, etc.

Intersecting lines: When two or more lines cross each other in a plane, they are called intersecting lines. We see these lines in the designs on handkerchiefs, graphs, chess boards, window nets and roads that cross each other.







My Activity

Take *rangoli* or soft soil/sand in a big dish or a tray to practice lines on it using a stick or with a finger. Make diagrams of wavy and intersecting lines using thick wool.

- Make the children practice wavy and intersecting lines. Make them use different mediums like coloured pencils, crayons, coloured pens or sketch pens for practice.